

Learning Location, Not Looking: A Placement-Memorization Audit and Ten-Episode Repair at the 30M Scale

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Abstract

Fixed-placement benchmarks let imitation policies learn *location* instead of *looking*. On LIBERO-Object we measure the preconditions exactly: the target’s start pose is bit-identical across all 50 shipped init states in 6 of 10 tasks, and the ten tasks between them use just **two** table positions, 22 cm apart. A policy that memorizes two coordinates can present as generalizing over ten.

We audit our own stack and find placement memorization at **four** layers: a calibrated expert whose offset constant encodes pose, an iteration-coupled selection loop, a grasp head whose regression prefers proprioception to pixels, and a place head that memorized a command→location map against a basket that never moves. Three are repaired, each by *data* rather than architecture: ten placement-teleported episodes, command coverage in training, and DAGger on the machine’s own deployment viewpoints. The released head scores 0.700 held-out (35/50 [0.56, 0.81]) and 0.520 randomized (26/50) where free per-tick regression scores 0/56; a third object crosses at 18/50 [0.24, 0.50] against a same-seed control at 3/50 (exact McNemar $p=6.1 \times 10^{-5}$). Twin control cells scope that verification from both sides: the behavioural certificate is joint in (*head*, *stack*)—a detector-stack rebuild inverts which head looks better.

The platform is 30.2 M parameters deployed with *no* language model—an open-vocabulary detector’s own text tower supplies every text embedding—which makes it the smallest surveyed language-conditioned policy under both size conventions at once. Nothing below is attributed to size.

Two annotation-free probes generalize past this artifact. Comparing two instructions’ detections on one frame exposes a grounding stage that *ignores* the instruction; swapping the instruction exposes one that *memorized* it. Applied here they show that object selection, approach and grasping run with no language input at all: this system’s entire language channel is one latched place coordinate.

All results are simulation, one task and one object per claimed cell. Seven pre-registered predictions were falsified and eight instruments retired at their own calibration gates; two proposed mechanisms were tested and retracted before reaching this paper.

1 Introduction

A policy that scores 0.700 on a manipulation benchmark has, on the face of it, learned to see, decide, and act. This paper is about how much of such a number can instead be *location*: constants—in a benchmark’s init files, in an expert’s calibration, in a learned regressor’s shortcut—that encode where the object always is, so that vision only ever gates an approach a memorized value finishes. On LIBERO-Object [11] we measure the preconditions exactly: the target object’s start pose is bit-identical across all 50 shipped init states in 6 of 10 tasks, and varies by only 0.25–0.58 cm (std) in the other four. *Across* tasks the structure is tighter still—the ten targets occupy just **two** table positions, 22.0 cm apart, five tasks each, agreeing to 0.0–0.7 mm within a group. A policy that memorizes two coordinates can present as generalizing over ten tasks. A benchmark with $\lesssim 1$ cm of within-task target variance and two positions between them admits position-memorizers, and we found them in every layer of our own system.

The vehicle is MicroVLA, a deliberately small language-conditioned vision-language-action stack: ~ 30 M parameters deployed, with a frozen open-vocabulary detector serving as the only vision *and* text encoder. The size is the paper’s setting, not its claim. Its methodological value is that a stack this small has no capacity to paper over systems defects, so every seam had to be instrumented—and the instruments are what this paper contributes.

Contributions.

1. **A placement-memorization audit and its verification method.** A forensically worked instance of shortcut learning [6, 7] on a fixed-placement benchmark, shown at four layers, with the corrected measured account of what LIBERO-Object pins (per-task table and shipped measurement script); and the probe-plus-behavioural-randomization protocol, motivated by the on-manifold limit of substitution probes and by a counterexample to probe-only certification. The protocol’s twin control cells sharpen its scope: behavioural randomization separates memorized from repaired heads under identical draws on the deployment stack (1/10 vs. 4/10), and fails

to separate them on a stack whose rebuild disables visual goals (6/10 vs. 0/10).

2. **A controlled policy-class result.** With perception, trunk, and corpus regime fixed, free per-tick regression scores 0/56 (upper 95% bound 6.4%) while goal-space supervision driving an engineered pick-and-place shell scores 4/10 dev, 7/10 held-out, 4/10 randomized—claimed strictly as a bundle, with the shell’s share measured on both stacks.
3. **Training–serving-skew instrumentation.** 29 worked defect instances at producer/consumer seams, split into disagreements catchable by parity testing and agreements-on-a-wrong-convention catchable only by provenance [3, 16].
4. **A ground-truth-free test for identity-blind grounding, and the negative result it produced.** Open-vocabulary detectors are typically prompted with a fallback chain, so a policy can appear language-conditioned while every instruction collapses to the same generic prompt. Comparing two *different* instructions’ selections on the same frame detects this without knowing which is correct—the property that let it succeed where six ground-truth-seeking instruments failed. Applied to our own stack it returns the same box for different objects (median centre distance 0.0001), so our best cell’s 0.700 is *not* explained by binding the named object. Its behavioural companion is equally cheap: **swap the instruction to a different object and rescore against the original task.** Neither needs annotation, both need only a second forward pass, and they fail differently—the first exposes a grounding stage that ignores the instruction, the second a stage that memorized it. The swap is what found our fourth memorization layer, and it works precisely where a fixed benchmark cannot: when every task shares a target, a memorized command→location map and a grounded one are behaviourally identical until the command is changed. Both ship as `eval/probes.py` with a test suite, and their verdicts refuse the three over-claims this campaign got burned by: too few co-detected frames returns INCONCLUSIVE rather than a rate; a discriminating result states that it cannot see whether the discrimination is *correct*; and an underpowered cell prints its own ceiling.

Both headline findings have one shape: apparent competence resting on a channel other than the claimed one—position where we claimed looking, shape where we claimed naming. The audit generalizes better than the artifact does.

Table 1: Deployed parameter ledger. “Deployed” counts every parameter loaded at inference, frozen or trained.

component	params
YOLO-World-S detector (vision <i>and</i> the only text encoder)	13.0 M
Tiny Recursive Model world model	9.97 M
fusion + drift + relational + planner trunk heads	7.0 M
goal heads (grasp 0.17 M + place 0.07 M) + learned gates	0.24 M
deployed total	30.2 M

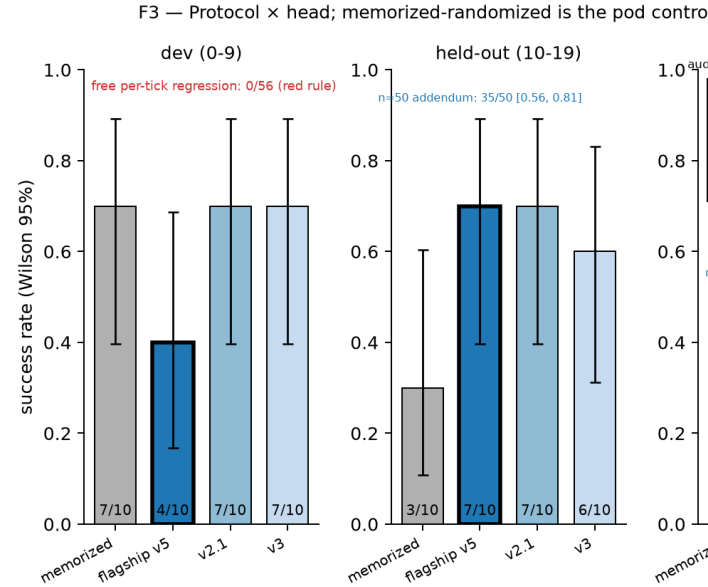


Figure 1: Protocol $\times$ head, Wilson 95% intervals. The memorized head leads on *dev* and collapses on held-out and randomized; the released head is flat across all three. The randomized panel’s inset is the audit-stack control, where the separation *inverts* (memorized 6/10, released 0/10) — it is never pooled with the deployment-stack bars, which is why the behavioural certificate is joint in (head, stack). Free per-tick regression, same trunk and corpus regime, scores 0/56.

2 Platform

No separate language model exists anywhere in the stack: the command is parsed into source/target phrases, and CLIP embeddings for (command, source, target) are harvested once per task from the detector’s own text tower. **The world model is causally inert in every nonzero number of this paper**—a persistence-TRM ablation reproduces the held-out cell exactly (0.700).

3 Results

Confirmed at power: held-out $35/50 = 0.700$ [0.56, 0.81]; randomized $26/50 = 0.520$ [0.39, 0.65]. Across a detector-stack rebuild the repaired head reproduces in all three protocols while the memorized head’s dev cell collapses $7/10 \rightarrow 2/10$ to its held-out floor: memorization is stack-coupled, *visual* grounding—which input the head reads—

Table 2: Protocol $\times$ head. All cells $k/10$ with Wilson 95% intervals; task 0, wrist camera. Audit-stack cells are never pooled with deployment-stack cells (§ stack finding).

protocol	memorized head	flagship (released)
dev (states 0–9)	7/10	4/10
held-out (states 10–19)	3/10	7/10
randomized ± 4 cm (same draws)	1/10	4/10
randomized ± 4 cm (audit stack)	6/10	0/10

is what survived.

4 Multi-object addendum

Four objects were attempted and three cross: 35/50, a second confirmed twice on never-scored seeds, and a third at **18/50** [0.24, 0.50] against a same-seed control at 3/50 (paired exact McNemar $p=6.1 \times 10^{-5}$). The fourth’s scripted teacher never reached a grasp, so no corpus exists for it; it is untested rather than failed.

For the third object we proposed nine mechanisms and **refuted eight by measurement**: generic-tail indiscriminability, box-centre spread, proposal scarcity, misbinding, position memorization, descent failure, binder quality, and uv-tail extrapolation. Two were our own and were retracted before reaching this paper. The object is notably *not* misbinding: a world-space probe finds it reaches the commanded object 6/6, failing by a systematic 6.9 cm *lateral* error (coherence 0.976) with height *better* than the object that works.

The surviving mechanism is appearance-side off-manifold drift, and it survives three tests. *Falsification*: the blocked object’s deployment embeddings sit $3.1\times$ further from its own corpus than the crossing object’s, and the second crossing object—predicted before it ran—has the smallest gap of all. *Circularity*: a wrong goal would itself produce unusual viewpoints, so we split by tick; drift is largest in the first 100, before the arm has acted, and shrinks 65%, while the crossing objects show the opposite rising trend. *Intervention*: fine-tuning the grasp head on the machine’s own viewpoints —labelled from the simulator’s object pose, which the corpus cannot supply for viewpoints the teacher never visited—cuts the deployed lateral error 56% ($p=0.020$) and produces the cell above. Collection drew seed 0 and evaluation seed 20, so no cell is scored on its own band.

The repair is **specific**, which is its best evidence. Applied to the object that already works it changes nothing: 28/50 vs. a 27/50 control, 21 discordant pairs split 11–10, $p=1.0000$. Three times the drift, six times the benefit; a third of the drift, none at all. A repair improving every object would be consistent with generic regularization; one appearing only where the measured defect is present is difficult to attribute otherwise. It repairs *covariate shift*, not absent labels, and only where viewpoints have actually drifted.

5 Related work

Size is comparable only under a stated convention, and the two in common use disagree about which system is smallest, because published counts routinely exclude frozen language encoders that inference nonetheless uses. We report both (Table 3). Under *trained parameters* the smallest published alternative is Octo-small’s 27 M transformer [15], and MicroVLA’s 17.2 M is $1.6\times$ below it; under *deployed* = every parameter loaded at inference it is RT-1’s 35 M [4], and MicroVLA’s 30.2 M is below that *even after granting RT-1 a free language encoder*—RT-1’s count excludes the Universal Sentence Encoder producing its instruction embedding, for which we cite no figure rather than estimate one. MicroVLA is thus smallest under both conventions simultaneously, the strongest form of the claim the evidence supports. The survey was run adversarially, against systems whose stated goal is minimal size: NanoVLA-S is 52 M trainable atop a frozen ResNet18 *and* a frozen BERT-base (~ 161 M loaded) [14], and LiteVLA reaches a Raspberry Pi on a 256 M SmolVLM backbone [10]. Both sit above MicroVLA on both axes, and NanoVLA is the convention gap in miniature—a paper named for being nano, reporting 52 M while loading ~ 161 M, because the frozen language encoder is not counted.

Table 3: Size under both conventions. † excludes a frozen language encoder the published count omits. Sizes only: capability, task breadth, and protocol are not comparable.

system	trained	deployed	language encoder
RT-2 [5]	55 B	55 B	internal (PaLI-X)
OpenVLA [9]	7 B	7 B	internal (Llama-2)
SmolVLA [18]	450 M	450 M	internal (SmolVLM2)
LiteVLA [10]	LoRA only	~ 256 M	internal (SmolVLM-256M)
NanoVLA-S [14]	52 M	161 M	BERT-base, frozen
Octo-small [15]	<u>27 M</u>	138 M	T5-base, 111 M frozen
RT-1 [4]	35 M	≥ 35 M†	USE, frozen, uncounted
MicroVLA	17.2 M	30.2 M	<i>none</i> —detector’s own tower

The claim is bounded and checkable: it ranges over language-conditioned VLA policies with published parameter counts that we surveyed, it is a statement about parameter counts alone, and it is refuted by exhibiting any surveyed-class system below 17.2 M trained or 30.2 M deployed. We cannot rule out unpublished or unsurveyed smaller systems. “Language-conditioned” here names the *reference class* — systems that accept a language instruction — and is not a claim about how much work that instruction does in ours: the decomposition above shows it reduces to one latched place coordinate. A reader who thinks that disqualifies the artifact from the class should read the size row as smaller still, not larger. The mechanism behind the gap is architectural rather than incidental: MicroVLA carries no separate language model at all, because the detector’s own text tower supplies every text

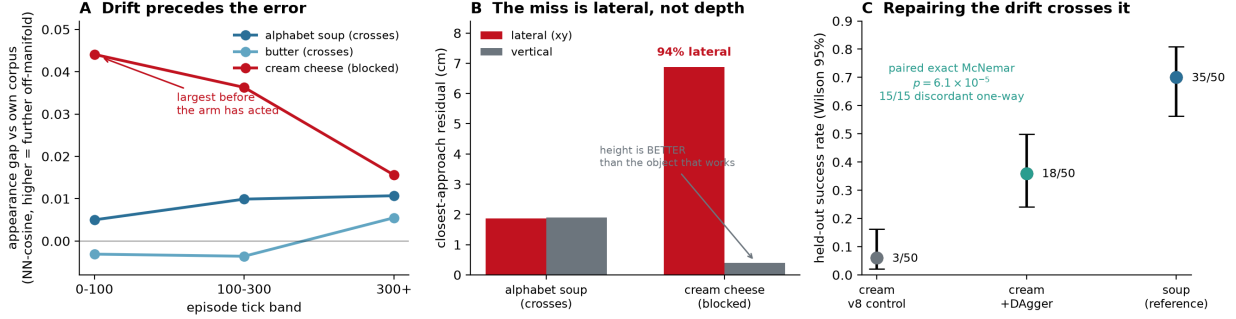


Figure 2: How the third object crossed. **(A)** The blocked object’s appearance drift is largest *before the arm has acted* and shrinks 65%, where both crossing objects show the opposite rising trend — the circularity check that retired two earlier instruments. **(B)** The miss is 94% lateral, with the blocked object’s *height* better than the crossing object’s. **(C)** The intervention, $n=50$ per arm on identical seeds: $3/50 \rightarrow 18/50$ (paired exact McNemar $p=6.1 \times 10^{-5}$), against the crossing object’s $35/50$.

embedding—so the frozen encoder that the other rows pay for twice (once in memory, once in the convention mismatch) does not exist here. Of MicroVLA’s 17.2 M trained parameters only 7.24 M are causally load-bearing, since the 9.97 M world model is inert in every nonzero number of this paper. No number here is comparable to community-protocol LIBERO scores (600 steps, $n=10$, wrist-only, scripted-expert corpora). Predicting *where* rather than *how* is the Transporter [20] and CLIPort [17] lineage, continued by keypoint interfaces over engineered primitives [8, 12] and, with both levels learned, $\pi_{0.5}$ [2]. Our audit is a worked manipulation instance of causal confusion and copycat shortcuts [1, 6, 19]; what we add is the audited counterexample to probe-only certification, the resulting probe-plus-randomization pairing, and the control cells that scope the pairing itself. Bootstrapping from scripted experts is established [13]; our delta is that the scripted teacher is itself audited, caught encoding placement and detector version, and repaired.

6 Claims, non-claims, limitations

Claimed: the pinning measurement in its scoped form; the released head’s cells above; free per-tick regression at 0/56; the de-memorization mechanism at the head level (attribution shift, varied-label validation, and a paired jitter ablation, 0/10 $\rightarrow$ 7/10 dev, exact McNemar/sign $p \approx 0.016$), scoped by the audit-stack control; 29 worked skew defects; the bounded size result of Table 3 (smallest of the surveyed systems under both the trained and the deployed convention simultaneously); and the identity-blindness of deployed role binding, together with its architectural corollary that no text embedding reaches the grasp head—claimed as a property of *this system*, and offered as a test other detector-grounded policies can run, not as a claim about them. **Not claimed:** any task-success contribution from the world model (causally inert

in every nonzero number); end-to-end learned motor control (the shell is engineered); head-level grounding from the randomized column alone, stack-free; object-level generalization beyond the addendum’s two crossed objects (four attempted, three crossed; the fourth’s scripted teacher never reached a grasp, so no corpus exists for it and it is untested rather than failed); any deployed binding-accuracy figure beyond the world-space nearest-body counts; physical deployment; any size claim beyond Table 3’s bounded one (surveyed systems, published counts, parameters only)—neither that smaller systems cannot exist, nor that small size caused any result here. **Limitations:** $n=10$ per cell with $n=50$ confirmations on two protocols; ten held-out states reused across five disclosed selection looks; machine constants inherited from a dev-tuned era; substitution probes are on-manifold; ± 4 cm randomization sits inside the shell’s ± 6 cm probe envelope and separates heads only where visual goals are live; the teacher’s calibration consumed privileged telemetry offline; single benchmark, single simulator.

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